

Joshua Lear

(309) 781-0254 | learjoshuab@gmail.com | <https://github.com/learjosh72> | <https://www.linkedin.com/in/joshua-lear>

EDUCATION

Purdue University	West Lafayette, IN
Bachelor of Science in Computer Science (Machine Intelligence & Security Concentrations)	May 2026
Minor: Psychology	
Cumulative GPA: 3.68	
Dean's List: Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026	
Semester Honors: Fall 2022, Spring 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026	
Relevant Coursework: Data Structure and Algorithms, Systems Programming, Data Mining and Machine Learning, Introduction to Artificial intelligence, Information Systems, Web Information Search & Management, Data Engineering in Python, AI Basics, Operating Systems	
Certifications: AWS Cloud Practitioner Essentials	

SKILLS

Programming Languages: Python, C, C++, Java, SQL, HTML, CSS, x86-64 Assembly, Bash Scripting
Technologies: AWS, Linux, Neo4J, MongoDB, MySQL, SQLite
Developer Tools: Vim, GitHub, VS Code, IntelliJ, PyCharm
Personal Skills: Communication, Problem-Solving, Critical Thinking, Leadership

PROJECTS

ERD Grade Predictor | *Python*

- Developed machine learning models to predict grading outcomes for Entity Relationship Diagram submissions.
- Applied information retrieval and similarity techniques including TF-IDF, graph similarity metrics, and k-nearest neighbors classification.
- Processed and analyzed structured diagram representations to evaluate semantic similarity between student submissions.
- Trained and evaluated predictive models on labeled grading datasets.

Shell Interpreter | *C, C++, lex, yacc*

- Developed a custom Unix-style shell capable of executing commands and handling scripting in Linux.
- Implemented parsing and command interpretation using lex and yacc.
- Integrated features including pipes, input/output redirection, and multi-process command execution.
- Applied operating systems concepts involving process control, system calls, and shell execution behavior.

Exam Manager | *Python, SQLite, HTML, Flask, SQLAlchemy, Bootstrap, Jinja*

- Developed a Flask-based web application for managing exams, student registration, and database-backed scheduling workflows.
- Designed relational database schemas and integrated SQLAlchemy ORM models for persistent data management.
- Implemented dynamic HTML templates and responsive UI components using Jinja and Bootstrap.
- Built server-side filtering, form handling, and user interaction logic for exam management operations.

WORK EXPERIENCE

Target | *Seasonal Tech Consultant*

June 2024 – August 2024

- Helped guests with any technical issues they may have
- Advised guests on purchasing decisions based on technical specifications
- Drove the technology business within the store

ACTIVITIES

IEEE Computer Society	Fall 2022-Spring 2023
Purdue University All American Marching Band	Fall 2023- Spring 2026
Boiler Brass Purdue Men's Basketball Pep Band	Fall 2022- Spring 2026